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Current status of the high-temperature kinetic models of silane: Part I. Pyrolysis

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ABSTRACT

The present work compares the performance of seven reaction models with respect to a large experimental dataset relevant to the high-temperature pyrolysis of both silane (SiH_4) and disilane (Si_2H_6). Their performances were established based on different validation criteria that account for the shape and the amplitude of the validation profile. Then, the model performances were quantified with a global error, which accounts for the experimental uncertainties. The most satisfactory model has a global error as low as 3.1 (i.e., meaning 3.1 times higher than the experimental uncertainty) and the highest fraction (74%) of criteria with a low error (< 2), while most of the models have large discrepancies with the validation dataset, global error near 8 and up to 110 for the less accurate model. The origins of these discrepancies are identified with reaction pathway and sensitivity analyses. Among the seven tested model, three main decomposition pathways are evidenced, including one more specific to the models presenting the lowest errors. Based on the global error values, the ability to reproduce all the experimental conditions, and the model analyses, the reaction pathways relevant to the high-temperature pyrolysis of silane and disilane are determined. In addition, the present study provides experimental and numerical guidance for the future developments of silicon hydride reaction models. The limited performance of most of the oldest reaction models may have a significant impact on our current understanding of the pyrolysis and oxidation kinetics of silane.

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1. Introduction

Silane and other small silicon hydrides (Si_xH_y) are involved in the production of a large number of products related to the semiconductor industry. They are used to manufacture insulation materials [1], silicon nanowires [2], solar panel elements [3], catalysts, and electronic devices. Unlike the hydrocarbon kinetics [4–12], the kinetics of silicon containing molecules have been less investigated. The need for accurate high-temperature reaction model for silicon hydrides, such as silane and disilane, recently regained interest with the development of high-temperature synthesis techniques for advanced nanomaterials, such as self-propagating high-temperature synthesis (SHS), flame spray pyrolysis (FSP), or combustion synthesis [13]. In addition to its application to mate-

rials science, the high-temperature pyrolysis of silicon hydrides is also of interest for risk assessment studies [14,15] and propulsion [16,17].

The first silane reaction models were developed for the high-temperature combustion by analogy with the methane combustion, from the pioneering work of Jachimowski and McLain [18], Hartman et al. [19], and Britten et al. [20]. Later, pyrolysis-specific reaction submodels started to be investigated in the 90's. A reaction model composed of 20 reactions was developed for silane and disilane (Si_2H_6) pyrolysis by Mick [21] and Woiki et al. [22]. This model was later updated by Petersen and Crofton [23]. In addition, several reaction models for silane oxidation kinetics, including pyrolysis submodels, were developed by Babushok et al. [24], Kondo et al. [25], Miller et al. [26], and Mével et al. [27–29]. Some of these oxidation models also include particles formation pathways, which are of interest for material synthesis processes, such as the chemical vapor deposition (CVD) processes [30]. Recently, new reaction models were developed, such as the updated reaction model of Petersen (also called PeOx, recently presented in [31,32]), the

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